Guidelines for Agronomists: Writing Effective Knowledge Templates for AI RAG Agents

Objective of These Guidelines

These guidelines will help agronomists write structured and optimized knowledge templates that can be **easily processed, chunked, and retrieved by an AI agent**. The AI agent will provide **personalized, time-sensitive, and context-aware recommendations** to farmers.

To ensure **accurate and efficient retrieval**, follow these writing rules. Each rule is designed to make the AI **retrieve the right information at the right time** without errors or confusion.

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1. Use Plain Text with Structured Headers and Bullet Points (No Tables)

✅ **Why?**

- AI models **understand text better than tables** because tables often lose structure when converted into embeddings.
- Using **headers and bullet points** ensures the information is stored and retrieved in **meaningful, topic-specific chunks**.
- Farmers' queries are often **context-based** (e.g., "What should I do if my soil is too acidic?"), and a **structured text format** allows the AI to provide an answer **without missing details**.

✅ **How to Apply This Rule**

- **DO:**
  - Write in **short paragraphs** with **clear section headers**.
  - Use **bullet points** for step-by-step instructions and lists.
  - **Group related information together** under a single heading.
- ❌ **DON'T:**
  - Use tables to present structured data (instead, write it as text).
  - Write long, dense paragraphs without subheadings.

✅ Example (Correct Way to Write It)

Soil Requirements for Coffee Robusta

- **Optimal pH:** 5.5 - 6.5
- **Soil Type:** Well-drained, loamy soil with high organic matter.
- **Common Deficiencies & Solutions:**
  - **Acidic soil (<5.5 pH):** Apply **agricultural lime**.
  - **Alkaline soil (>6.5 pH):** Use **elemental sulfur**.
  - **Sandy soil:** Increase organic matter with compost and mulch.

❌ **Example (Incorrect - Using a Table)**

Soil Type	Optimal pH	Deficiency	Solution
Loamy	5.5 - 6.5	None	Maintain good organic matter
Acidic	<5.5	Nutrient lockout	Apply lime

🚫 **Why is this incorrect?**

- AI might **retrieve only one row** from the table instead of the full context.
- AI cannot **reason about** cause-and-effect relationships across different table rows.

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2. Keep Each Section Self-Contained (Modular Structure)

✅ **Why?**

- AI retrieves **small chunks of information at a time**. If knowledge is spread across different sections, the AI **might retrieve an incomplete answer**.
- Writing in **standalone sections** ensures that the AI can **return the right answer even if it only retrieves one part of the document**.

✅ **How to Apply This Rule**

- Each **header should introduce a complete idea**.
- Do **not reference previous sections** (e.g., don't say "As mentioned above"—repeat the key point).
- If a section depends on another, **briefly summarize** the related information instead of assuming the AI remembers it.

✅ Example (Correct Way to Write It)

How to Prevent Coffee Leaf Rust

- **Symptoms:** Yellow-orange powder spots on leaves, leaf drop.
- **Prevention:**
  - Use **rust-resistant coffee varieties**.
  - Prune trees to **increase airflow** and reduce moisture buildup.
  - Apply **copper-based fungicides** before the rainy season.
- **Treatment if Already Infected:**
  - Remove and burn infected leaves to prevent spread.
  - Apply **systemic fungicides** if more than **10% of the plant is affected**.

❌ Example (Incorrect - Not Self-Contained)

How to Prevent Coffee Leaf Rust

- See the "Disease Management" section above for prevention methods.

🚫 **Why is this incorrect?**

- The AI **may not retrieve the referenced section**, leading to **incomplete or missing information**.

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3. Use Simple, Direct Language (Avoid Technical Jargon)

✅ **Why?**

- Many farmers **do not have a technical background**.
- AI models work best with **clear, simple language** that does not require interpretation.
- If a term **must** be technical, **define it immediately**.

✅ **How to Apply This Rule**

- **Use simple words**: Instead of "photosynthesis efficiency," say "how well the plant makes food from sunlight."
- **Define technical terms** in the same sentence where they appear.

✅ Example (Correct Way to Write It)

How to Improve Soil Fertility for Coffee Trees

- **Compost:** Improves soil **structure and nutrients**. Apply **5 kg per tree, twice a year**.
- **Nitrogen (N):** Helps plants grow leaves. If leaves look **pale or yellow**, apply compost or organic fertilizer.
- **pH Balance:** Soil should not be **too acidic** or **too alkaline**. Test with a pH kit and adjust if needed.

❌ Example (Incorrect - Too Technical)

Soil Fertility Enhancement Using Organic Amendments

- Organic matter improves cation exchange capacity (CEC), leading to enhanced nutrient bioavailability.
- Excessive nitrogen volatilization occurs if soil buffers are inadequate.

🚫 **Why is this incorrect?**

- Farmers (and even some agronomists) **may not understand complex terminology**.
- AI might **misinterpret the technical explanation**, leading to confusing or inaccurate responses.

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4. Include "When & Where" Context (Make Recommendations Time-Specific)

✅ **Why?**

- Farmers ask **time-sensitive questions** (e.g., "What should I do in February?" or "What fertilizer should I apply before the rainy season?").
- AI works better when recommendations include **specific timeframes** and **regional conditions**.

✅ **How to Apply This Rule**

- Mention **seasonal timing** ("Apply in the dry season before flowering").
- If a **recommendation depends on a location**, specify the region (e.g., "In Vietnam, apply before monsoon rains").

✅ Example (Correct Way to Write It)

Fertilization Schedule for Coffee Robusta

- **Before Flowering (March - April):** Apply phosphorus-rich fertilizers like **bone meal** to encourage root growth.
- **During Berry Formation (May - June):** Apply **potassium (K)** to improve bean quality.
- **After Harvest (December - January):** Add compost to restore soil nutrients.

❌ Example (Incorrect - No Timing Context)

- Apply fertilizers regularly to keep plants healthy.
- Use phosphorus for roots and potassium for fruits.

🚫 **Why is this incorrect?**

- AI **won't know when** to apply the fertilizers.
- Farmers may **use the wrong fertilizer at the wrong time**.

Final Writing Checklist for Agronomists

✅ **Use plain text with headers and bullet points (No tables).**
✅ **Write each section as a self-contained chunk (No references to other sections).**
✅ **Use simple, direct language (Avoid technical jargon or define it immediately).**

✅ **Include “When & Where” context for time-sensitive recommendations.**
✅ **Write in a way that a farmer can understand without expert knowledge.**

🚀 **By following these guidelines, the AI will retrieve and generate the most accurate, context-aware, and actionable responses for farmers!**

Optimizing Documents for Semantic Chunking by an AI Agent

To ensure that an AI agent efficiently processes and retrieves relevant information from a document, the document must be structured in a way that supports semantic chunking—breaking text into meaningful, contextually complete segments.

Below are best practices for structuring documents optimized for semantic chunking in AI-driven retrieval systems (e.g., RAG models, vector search, knowledge retrieval).

1. Use a Hierarchical Structure

AI agents perform better when documents have a clear hierarchical structure with logical segmentation. Use headings and subheadings to create distinct chunks.

Example Structure:

Main Topic

Subtopic 1

Key Concept A

Supporting Detail

Subtopic 2

Key Concept B

Supporting Detail

Best Practices:

- Use markdown syntax and stick to simple formatting - no fancy formatting
- Each section should be logically independent to allow retrieval without needing excessive surrounding context.
- AI models favor consistent patterns, so keep formatting uniform.

2. Keep Chunks Concise but Complete

AI semantic chunking algorithms work best with self-contained yet concise segments.

Recommended Chunk Lengths:

Content Type	Optimal Token Range
Key definitions	50-100 tokens
Concept explanations	100-300 tokens
Process descriptions	200-500 tokens
Case studies	300-800 tokens

Example (Bad Chunking – Too Large):

"Regenerative agriculture improves soil health by increasing organic matter. Organic matter enhances microbial activity, improves soil structure, and increases water retention. One common method is cover cropping, which involves planting crops like legumes and grasses to fix nitrogen and prevent erosion. Another technique is compost application, where decomposed organic material is added to enrich the soil. Reduced tillage is also beneficial as it minimizes soil disruption, preserving beneficial organisms and preventing carbon loss."

Example (Good Chunking – Modular & Retrievable):

Definition: Regenerative agriculture is a farming approach that enhances soil health by improving organic matter and biodiversity.

Method 1 – Cover Cropping: Planting legumes and grasses to fix nitrogen and reduce erosion.

Method 2 – Compost Application: Using decomposed organic matter to enrich soil fertility.

Method 3 – Reduced Tillage: Minimizing soil disturbance to preserve beneficial microbes and prevent carbon loss.

6. Embed Inline Definitions & FAQs

AI models retrieve short answers effectively from FAQs and definitions.

Example (Good Format - Inline Definitions & FAQs)

What is Cover Cropping?

Cover cropping is the practice of planting specific crops (e.g., legumes, grasses) to improve soil health, prevent erosion, and fix nitrogen.

How does Cover Cropping Improve Soil?

1. Nitrogen Fixation: Legumes like clover fix nitrogen, reducing fertilizer dependency.
2. Erosion Prevention: Deep-rooted grasses protect topsoil.

3. Soil Microbiology Boost: Organic matter increases microbial diversity.

7. Use Clear, Natural Language

- Avoid ambiguity and jargon unless clearly defined.
- Ensure consistency in terminology (e.g., always use “carbon sequestration” instead of switching between “carbon storage” and “CO_2 capture”).
- Favor concise, clear sentences.

Example (Bad - Ambiguous & Unstructured)

“Soil amendments like biochar and compost affect soil differently. The former increases aeration, while the latter enhances microbial activity.”

Example (Good - Clear & Structured)

- Biochar: Improves aeration and water retention.
- Compost: Boosts microbial life and provides slow-release nutrients.

Final Takeaways:

✅ Use clear headings & subheadings to create logical chunking.

✅ Keep chunks concise but complete (~100-500 tokens).

✅ Ensure natural, jargon-free language for better AI understanding.

✅ Store chunks in a vector-friendly format for semantic search.

By following these principles, you ensure that AI agents can efficiently parse, chunk, retrieve, and respond to queries with high accuracy.